Jianshen **Liu**

Principal Engineer · Agentic AI Storage & Distributed Systems Expert

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Summary

Principal engineer with 15+ years of experience building low-latency storage, data-path, and AI infrastructure for large-scale inference and agentic AI workloads. Currently leading Agent Memory Storage and File-over-Memory initiatives at Huawei Technologies Canada, enabling GPU-initiated I/O to remote storage and improving vLLM inference performance by 80%. Delivered an xPU data-direct accelerator reaching 2.9 GB/s, about 25% faster than NVIDIA BaM. Deep expertise in **C++**, **CUDA**, **Linux kernel**, **DPDK**, **SPDK**, **Apache Arrow**, SmartNIC/DPU systems, and distributed storage performance engineering, backed by 6 peer-reviewed publications on computational storage and embedded data services.

Skills

AI & Storage	Agent Memory Storage, File-over-Memory, vLLM, AiWS, GPU I/O, xPU Data Direct, Storage Offload
Systems	Distributed Storage, Linux Kernel, eBPF, SPDK, DPDK, Ceph, Arrow, NVMe, SmartNIC/DPU, PCIe
Programming	C++, CUDA, Python, Java, Bash, SQL, PHP, JavaScript, LaTeX
Cloud & DevOps	AWS, GCP, Docker, Kubernetes, Jenkins, Git, CI/CD, Bazel, Maven, Jupyter
Languages	English, Cantonese, Mandarin

Work Experience

Huawei Technologies Canada Co., Ltd. — *Markham, Canada*

Principal Engineer — *Apr 2025 – Present*

- Lead Agent Memory Storage implementation enabling GPU-initiated I/O directly to remote storage, reducing data-path overhead for agentic AI and inference workloads
- Drive end-to-end File-over-Memory delivery from research prototype through platform-team technology transfer
- Boosted vLLM inference performance by 80% by profiling end-to-end bottlenecks and optimizing the File-over-Memory data path
- Established code-review and quality standards for storage services, reducing validation cycles and improving service stability

Huawei Technologies Canada Co., Ltd. — *Markham, Canada*

Senior Engineer — *Aug 2023 – Mar 2025*

- Engineered xPU data-direct accelerator reaching 2.9 GB/s at 73% PCIe Gen3 utilization, 25% better than NVIDIA BaM
- Reduced File-over-Memory latency by 90% through protocol optimization between client library and manager
- Developed prefill/decode (PD) disaggregation for vLLM and session management for AI Workload Scheduler (AiWS), improving infrastructure support for scalable inference workloads
- Mentored 6+ engineers on distributed-system design, storage data paths, and I/O optimization

University of California, Santa Cruz — *Santa Cruz, USA*

Graduate Student Researcher — *Jan 2016 – Jul 2023*

- Researched SmartNIC/DPU and computational-storage architectures for in-transit data processing, embedded data services, and HPC storage systems at Sandia National Laboratories
- Implemented *Bitar*, a C++ library using Apache Arrow and DPDK, accelerating data compression >8x with BlueField-2 DPU
- Published 6 peer-reviewed papers on composable data services and storage offload; work featured in The Next Platform and Data Centre Dynamics
- Contributed to Ceph and Apache Arrow open-source projects, strengthening distributed-storage and columnar-data-system expertise
- Developed *no_fscache*, a Linux kernel module addressing page cache pollution issues in storage-system benchmarking

KIOXIA America Inc. — *San Jose, USA*

Strategic Marketing Engineer Intern — *Jun 2020 – Sep 2020*

- Quantified >10x performance speedup for SQL query offload to SSDs using MariaDB, Docker, and TPC-H benchmarks
- Collaborated across product and engineering teams to identify I/O performance issues in NUMA systems

Samsung Semiconductor — *San Jose, USA*

Performance Analysis Engineer Intern — *Jul 2019 – Sep 2019*

- Built distributed testing framework for Samsung Network KV device performance evaluation
- Created eBPF and kprobe observability tools, identified storage bottlenecks, and boosted write performance by 60%

Wumii Technology — *Shenzhen, China*

Android Development Team Lead — *Dec 2012 – Jun 2014*

- Led design and code reviews for a top-25 reading recommendation app in China
- Implemented energy-efficient Paho MQTT notification module in Java; 35% latency improvement over commercial solutions

Wumii Technology — *Shenzhen, China*
Senior Software Engineer — *Jun 2010 – Dec 2012*

- Architected first news recommendation system in China using Spring, Hibernate, and Google Closure Compiler
- Launched multi-platform recommendation platform (Java, PHP, JavaScript) serving 200M monthly active users

Education

University of California, Santa Cruz — *Santa Cruz, USA*
Ph.D., Computer Science and Engineering — *2016 – 2023*
Thesis: *Composable Data Services for Embedded Systems*; GPA: 3.96/4.00
Committee: Carlos Maltzahn (Chair, UCSC), Scott A. Brandt (UCSC), Peter Alvaro (UCSC), Craig D. Ulmer (Sandia National Laboratories)
Funding: DOE Office of Science/ASCR (FWP 20-023266); DOE NNSA (DE-NA0003525).

South China Agricultural University — *Guangzhou, China*
B.S., Information and Computing Science — *2006 – 2010*
GPA: 3.45/4.00

Honors & Awards

May 2023	**Best Paper Award**, 2nd Workshop on Composable Systems (COMPSYS '23)	*Florida, USA*
Sep 2022	**Outstanding Student Paper**, 26th Annual IEEE High Performance Extreme Computing (HPEC '22)	*Virtual Conference*
Oct 2009	**National Encouragement Scholarship**, South China Agricultural University	*Guangzhou, China*
Oct 2009	**The Second Prize Scholarship**, South China Agricultural University (top 10%)	*Guangzhou, China*
Oct 2008	**State Grants**, South China Agricultural University	*Guangzhou, China*
Oct 2008	**The Second Prize Scholarship**, South China Agricultural University (top 10%)	*Guangzhou, China*

Publications

Opportunistic query execution on smartnics for analyzing in-transit data
Jianshen Liu, Carlos Maltzahn, Craig Ulmer
2023 IEEE High Performance Extreme Computing Conference (HPEC), 2023

Extending Composable Data Services into SmartNICs
Craig Ulmer, Jianshen Liu, Carlos Maltzahn, Matthew L. Curry
2nd Workshop on Composable Systems (COMPSYS '23), Co-located with IPDPS 2023, 2023, Florida USA

Processing Particle Data Flows with SmartNICs
Jianshen Liu, Carlos Maltzahn, Matthew Leon Curry, Craig Ulmer
26th Annual 2022 IEEE High Performance Extreme Computing (IEEE-HPEC 2022), 2022, Virtual Conference

Performance Characteristics of the BlueField-2 SmartNIC
Jianshen Liu, Carlos Maltzahn, Craig Ulmer, Matthew Leon Curry
Sandia National Lab. (SNL-CA), Livermore, CA (United States); Sandia National Lab. (SNL-NM), Albuquerque, NM (United States) (May 2021). 2021

Scale-out Edge Storage Systems with Embedded Storage Nodes to Get Better Availability and Cost-Efficiency At the Same Time
Jianshen Liu, Matthew Leon Curry, Carlos Maltzahn, Philip Kufeldt
3rd USENIX Workshop on Hot Topics in Edge Computing (HotEdge'20), 2020, Virtual Conference

MBWU: Benefit Quantification for Data Access Function Offloading
Jianshen Liu, Philip Kufeldt, Carlos Maltzahn
International Conference on High Performance Computing (HPC-IODC 2019), 2019, Springer, Cham

Presentations & Posters

Experience of using the BlueField-2 DPU
Presented at Sandia National Laboratories (May 2022). 2022

Processing Particle Data Flows with SmartNICs
Presented at UC Santa Cruz Open Source Symposium (Sept. 2022). 2022

Eusocial Storage Devices
Presented at KIOXIA America Inc., Seagate Technology, and Fujitsu America Inc. (2020). 2020

MBWU (MibeeWu): Quantifying benefits of offloading data management to storage devices
Poster Session at 17th USENIX Conference on File and Storage Technologies (FAST'19) (Feb. 2019). 2019

Quantifying benefits of offloading data management to storage devices
Poster Session at 2019 OCP Global Summit Symposium (Mar. 2019). 2019

Certifications

Jan 2022	**Cruzhacks Mentor Certificate**, CruzHacks	*Santa Cruz, USA*
Dec 2020	**Graduate Student Professional Communication Certificate**, University of California, Santa Cruz	*Santa Cruz, USA*